YAHYA MENKARI

AI Engineer | Machine Learning | Data Engineering | RAG Systems

Arabic (Native) | French | English

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🌐 **Portfolio: gautax.github.io/yahya-portfolio**

PROFESSIONAL SUMMARY

Big Data & Artificial Intelligence Engineer with hands-on experience in Machine Learning, Data Engineering, Retrieval-Augmented Generation (RAG), and Graph Databases. Developed AI-powered applications using Neo4j, Qdrant, Gemini, Google Cloud, and modern ML techniques. Currently conducting machine learning research on weather forecasting at the Technical University of Sofia.

EDUCATION

International University of Rabat (UIR), Morocco

- M.Eng. in Computer Science – Big Data & Artificial Intelligence (2021 – 2026)

Technical University of Sofia, Bulgaria

- Exchange Program – IT for Business Management (2025 – 2026)

PROFESSIONAL EXPERIENCE

Machine Learning Research Intern | Technical University of Sofia, Bulgaria [View in Github]|
Feb 2026 – Jun 2026

- Developed an end-to-end weather forecasting pipeline using **43,000+ hourly observations** collected over a **5-year period**.
- Designed preprocessing and feature engineering workflows, expanding the dataset **from 19 to 43 predictive features** through cyclical encodings, lag variables, and rolling statistics.
- Implemented and benchmarked **XGBoost, GRU, CNN-LSTM, CNN-GRU, and TCN models** for 24-hour-ahead temperature forecasting.
- Optimized XGBoost performance, achieving **MAE 2.34, RMSE 3.05, and R^2 0.91** on unseen test data.

AI Research Intern | Archipel Cognitive, Morocco | Jul 2024 – Sep 2024

- Implemented sampling techniques (reservoir, weighted, stratified) and labeling functions in Python to simulate **ML data pipelines**
- Researched and documented **RAG system architecture** including **streaming ingestion**, CDC-based data capture, and **vector store** integration using **LlamaIndex, LangChain, and Qdrant**
- Studied production ML system design covering **reliability, scalability, maintainability, and model evaluation methods** across 6 chapters of Designing ML Systems

TECHNICAL SKILLS

- **Programming:** Python, SQL, Java, C
- **Data & AI:** Machine Learning, Deep Learning, Time-Series Forecasting, RAG, Agentic AI, Semantic Search, Knowledge Graphs
- **Databases:** MySQL, MongoDB, Neo4j, Qdrant (Relational, NoSQL, Graph & Vector Databases)
- **Tools & Platforms:** Scikit-Learn, XGBoost, CrewAI, Docker, Power BI, GCP, Azure

PROJECTS

Graph-Based AI Symptom Checker (*View GitHub*)| Neo4j, Qdrant, RAG, PubMed

- Built a medical knowledge graph and semantic retrieval system integrating Neo4j, Qdrant, and PubMed data.
- Implemented Retrieval-Augmented Generation (RAG) pipelines to improve contextual medical information retrieval.

Medical AI Assistant with Specialist Referral System(*View GitHub*) | CrewAI, Gemini, Streamlit

- Developed a multi-agent healthcare assistant automating specialist referral, report generation, and information retrieval workflows.

AI-Driven Prescription Reader(*View Github*) | Google Cloud Vision, Firestore

- Built an OCR-based prescription analysis system with multilingual medication validation and recommendation features.